# Fertility, Housing, and Population Dynamics

## From a stationary closure to a dated transition

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# Today: from the population closure to an equilibrium transition

**Last time:** we focused on the missing population closure:

$$\text{births today} \longrightarrow \text{new households later} \longrightarrow \text{housing demand and prices.}$$

**Today:** I try to make that idea an equilibrium object:

- build the simplest model with a well-defined steady state;
- separate the immediate response to a fertility-preference shock from the later demographic adjustment;
- add the same population accounting to the existing quantitative model;
- ask whether the 2007–2023 economy can be interpreted as a point along such a transition.

**The update is small: the same household architecture, a population closure, and one nested first-child housing term.**

# A toy economy

To fix ideas, let's look at a very simplified economy with young and old households:

- $Y_t$ is the number of young households at date $t$; $O_t$ is the number of old households.
- Total adult-household population is $S_t = Y_t + O_t$.
- A young household chooses consumption, housing services, and $n_t$ **household children**. Next period it becomes old.
- In this unitary-household model, each household child becomes one young household in the next generation.
- Both generations purchase housing services in the same market at price $p_t$.

The cohort law is therefore

$$Y_{t+1} = n_t Y_t, \qquad O_{t+1} = Y_t,$$

so $n_t = 1$ is replacement.

**Focus of the toy model:** fertility, cohort replacement, and the housing-services market. The quantitative model adds tenure choice, mortgages, saving, and bequests.

# Household choices and the housing market

A young household earns $y$ and chooses consumption $c$, housing services $h$, and household children $n$:

$$u(c, h, n) = c + \alpha \log h + v \log n, \qquad c + ph + \big[q + a(p + \omega)\big] n = y.$$

Here $\alpha$ values adult housing, $v$ values children, and each household child requires $a$ units of housing plus $q$ other goods. The term $\omega$ is the extra cost of accessing child-suitable housing.

The interior choices are

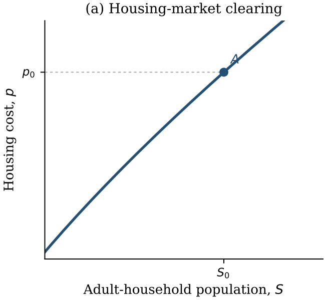
$$n(p) = \frac{v}{q + a(p + \omega)}, \qquad h_Y(p) = \frac{\alpha}{p} + an(p).$$

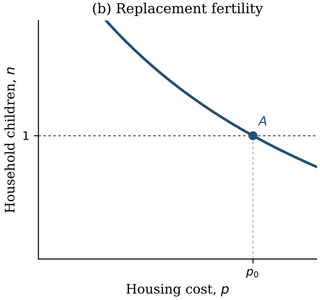
An old household has utility $c_O + \bar{h}_O \log h_O$, so its housing demand is $h_O(p) = \bar{h}_O / p$. Housing supply and market clearing are

$$H^s(p) = \bar{H} p^{\varepsilon}, \qquad Y_t h_Y(p_t) + O_t h_O(p_t) = H^s(p_t).$$

The parameter $\bar{H}$ scales housing supply and $\varepsilon$ is its price elasticity.
**A higher housing cost reduces both housing demand and fertility. Cohort sizes then feed back into the same market-clearing price.**

# Stage 1: the initial steady-state equilibrium




A steady state is a constant triple $(Y^*, O^*, p^*)$. It requires

$$n(p^*) = 1, \qquad Y^* = O^*.$$

Replacement selects

$$p_0 = \frac{v_0 - q}{a} - \omega.$$

Housing clearing then selects $S_0 = S^*(p_0)$, where

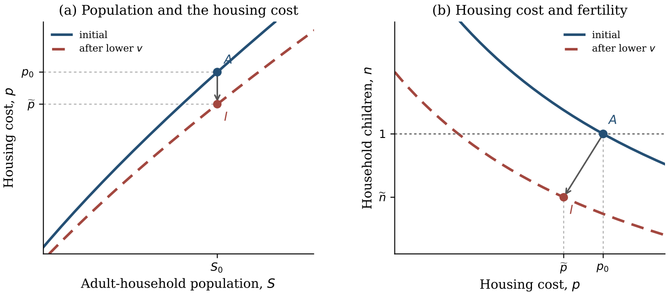
$$S^*(p) = \frac{2\bar{H}p^{1+\varepsilon}}{\alpha + \bar{h}_O + ap}.$$

Point $A$ is the initial steady state.

Along the steady-state schedule, direct differentiation gives

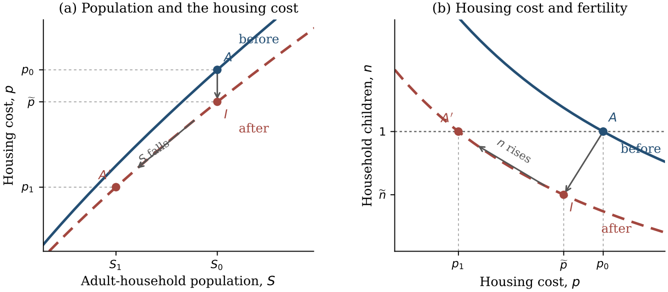
$$\frac{dS^*}{dp} = \frac{2\bar{H}p^{\varepsilon}\left[(1+\varepsilon)(\alpha + \bar{h}_O) + \varepsilon ap\right]}{(\alpha + \bar{h}_O + ap)^2} > 0 \qquad \text{for } \varepsilon \geq 0.$$

# Stage 2: the impact response




- **Shock:** $v_0 \to v_1 < v_0$.
- **Inherited cohorts:** $Y_t = Y_0$, $O_t = O_0$, $S_t = S_0$.
- **Housing cost:** lower child-related demand gives $p_0 \to \widetilde{p}$.
- **Fertility:** $\widetilde{n} = n(\widetilde{p}; v_1) < 1$.
- **Point $I$:** the impact equilibrium with inherited cohort sizes.

# Stage 3: demographic adjustment




- **Below replacement at $I$:**

$$Y_{t+1} = \widetilde{n} Y_t < Y_t.$$

- **Cohorts:** smaller young cohorts eventually reduce $S$.
- **Housing cost:** lower demand moves $\widetilde{p} \to p_1$.
- **Fertility:** lower $p$ raises $n$ along the red curve.
- **Point $A'$:** $n = 1$, $p_1 < p_0$, and $S_1 < S_0$.

**The price response cushions the demographic contraction: falling $p$ raises $n$, and the larger entrant cohorts slow the subsequent fall in both $S$ and $p$.**

# Below-replacement fertility need not lower population immediately

Let $n_t = n(p_t)$ be household children per young household and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $n_t < 1$ whenever

$$\mu_t < n_t < 1.$$

**A numerical example**

$$100 \text{ young} + 70 \text{ old} \quad \longrightarrow \quad 80 \text{ young} + 100 \text{ old}$$

$n_t = 0.8 < 1$, but total households rise from 170 to 180.

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1 - \mu_t)h_O(p_t) > (1 - n_t)h_Y(p_t).$$

**U.S. interpretation:** inherited cohorts can keep population and housing costs rising even after fertility falls below replacement.

# The quantitative model keeps the existing household problem

At date $t$, households differ in

age, wealth, income, location, tenure, house size, fertility history, and children at home.

They make the existing fertility, tenure, housing, saving, and moving choices. Let $G_t$ record the number of households at every combination of these characteristics.

The population closure carries that distribution forward in levels:

$$G_{t+1} = \text{surviving households after their choices} + \text{new entrants.}$$

**The change is that cohort masses now evolve in calendar time instead of being normalized back to a stationary distribution.**

# Births and the housing market close the aggregate transition

Two equations connect current choices to future population and current prices:

$$E_{t+5} = \frac{b_t}{2.1}, \qquad H^s(p_t) = D(G_t, p_t).$$

- Adjusted births $b_t$ generate entrant households twenty years later.
- The factor 2.1 converts child units into future household units.
- The price $p_t$ clears housing demand from the current distribution $G_t$.

Within the household problem, the first child can create a discrete housing need:

$$\bar{h}(m) = \bar{h}_1 \mathbf{1}\{m > 0\} + \bar{h}_c m,$$

where $m$ is children at home. The parameter $\bar{h}_1$ captures the first-child jump, while $\bar{h}_c$ captures the additional space required per child.

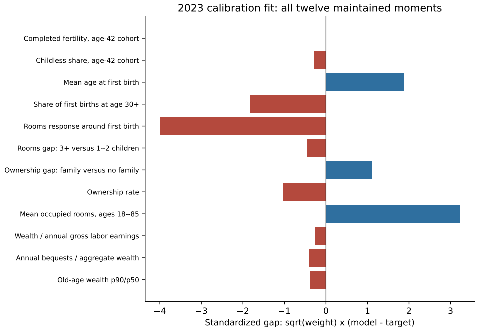
**Housing costs shape fertility; births change future cohort masses; those cohorts feed back into housing costs.**

# We calibrate an economy moving from 2007 to 2023

1. **Construct an old steady state.** Choose the initial fertility preference so completed fertility is 2.1. This is a replacement normalization, not a measured 2007 target.
2. **Construct the 2007 state.** Reweight the old distribution only by observed 2007 age shares, preserving wealth, tenure, and fertility states within each age.
3. **Run the economy to 2023.** Feed in Census household totals and ACS age shares. Let the fertility preference decline linearly, and estimate ten household-and-supply parameters and the total preference decline using twelve maintained moments, evaluated mainly in 2023.
4. **Check the path.** Compare untargeted model movements with period fertility, birth timing, ownership, rooms, house prices, and rents.
5. **Continue after 2023.** Hold preferences fixed and let the inherited household distribution and birth pipeline evolve.

**The population and age bridge are inputs, not successes of the model. The 2007 economy is a dated initial condition, not literally the old steady state.**

# The 2023 fit has two important housing-size misses




**Loss:** 36.099. Eleven parameters and twelve moments.

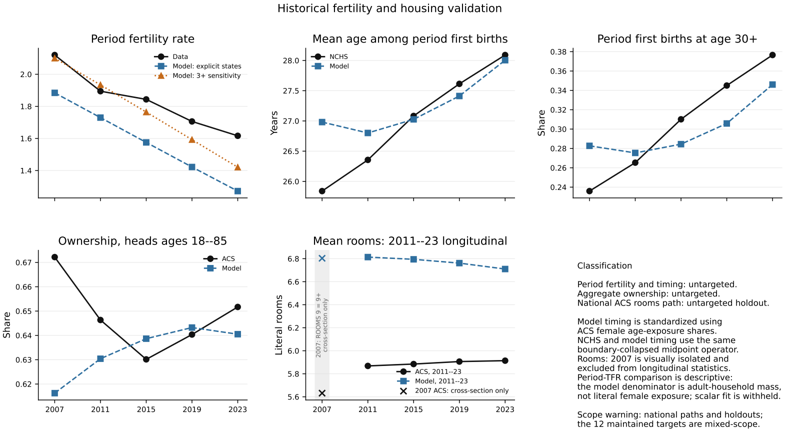
**Main misses**

- First-birth rooms: 0.380 versus 0.720
- Mean rooms: 6.710 versus 5.780

They contribute 26.30 of the total loss.

The first-child floor helps, but increasing it further worsens mean rooms and ownership. The PSID target remains provisional because its prepath is not flat.

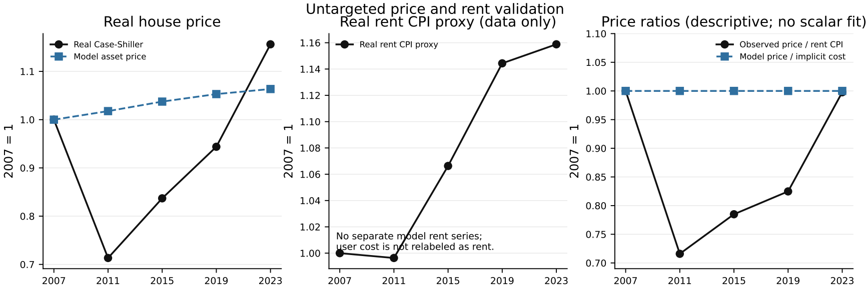
# Historical validation: close timing, weak housing fit




**Reading the figure:** timing is close by 2023; the model misses the ownership cycle and rooms trend, and its fertility decline is too steep.

National series are holdouts; maintained targets use mixed samples and vintages.

# The model does not reproduce the house-price cycle

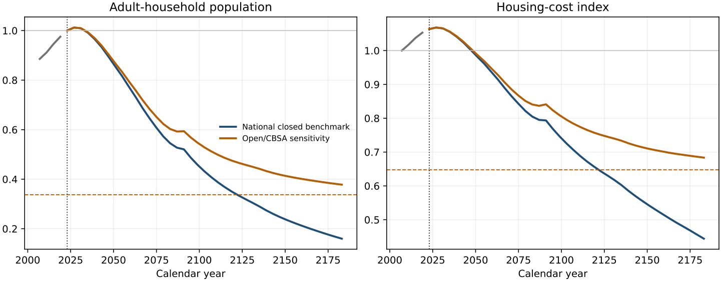



- The model cannot reproduce the 2007–2011 house-price collapse and later rebound.
- With temporary-equilibrium expectations and a fixed user-cost relation, it has no independent price-to-rent dynamics.

**The exercise should therefore be read as a demographic-housing transition, not as a complete account of the post-2007 housing cycle.**

Observed house prices: real Case-Shiller. The rent CPI is shown only as a descriptive proxy, not as the model's user cost.

# After 2023: the same state under two entry closures




Both paths start from the same calibrated 2023 distribution and birth pipeline, with fertility preferences held fixed.

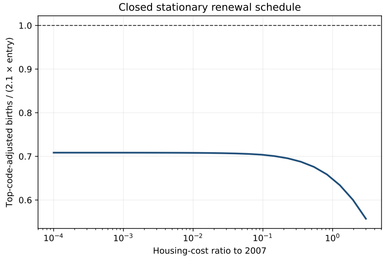
**Closed benchmark:** no outside entry; future entrants come only from earlier births.

**Open sensitivity:** the old outside flow $M$ and retention rate $\rho$ remain fixed. The outside-origin share can change.

The inherited twenty-year birth pipeline makes both paths peak slightly in 2027. Afterward, the closed economy contracts much faster.

Controlled no-policy equilibrium experiments, not forecasts.

# The fitted closed economy has no root on the audited grid




25-point grid, $P/P_0 \in [10^{-4}, 3]$; not a global theorem.

At each stationary price $P$, let $E(P)$ be entrants and $B(P)$ the future households generated by births. A positive closed steady state requires $B(P)/E(P) = 1$.

Instead, over the audited grid,

$$0.557 \leq B(P)/E(P) \leq 0.708 < 1.$$

Even almost free housing does not reproduce the entrant cohort.

**Therefore:** the closed continuation is a finite-horizon experiment, not a transition between two positive steady states.

# Can a stronger fertility response restore the root?

Starting from the immediately preceding calibration point, I change both fertility-logit noise scales by a common factor, re-normalize the old economy to replacement, and re-estimate the 2007–2023 preference decline so 2023 completed fertility remains on target.

| Logit-scale factor | 2007–2023 loss | Maximum $B(P)/E(P)$ |
|---|---|---|
| 0.25 | 1767.9 | 0.285 |
| 0.50 | 323.8 | 0.645 |
| 0.75 | 69.0 | 0.698 |
| 1.00 | 38.2 | 0.713 |
| 1.25 | 48.2 | 0.719 |
| 1.50 | 67.2 | 0.723 |
| 2.00 | 104.0 | 0.726 |

**Result:** none of the seven audited factors produces a closed steady-state root. The best transition fit remains near the original slope.

This rules out a simple common rescaling of the existing fertility response. It does not rule out a different fertility mechanism or a different population closure.

# Where this leaves the paper

- **The theoretical contribution is compact.** The existing housing–fertility mechanism is embedded in a population closure. Lower fertility eventually reduces population and housing costs, while inherited cohorts can make both rise initially.
- **The quantitative exercise is a dated transition.** We initialize in 2007, fit the maintained 2023 moments, and keep imposed demographic inputs separate from historical holdouts.
- **The present fit is informative but incomplete.** Several 2023 moments fit well, but housing quantities, the ownership cycle, and the house-price path do not.
- **The equilibrium problem is substantive.** The current fertility response is too weak to restore replacement at any audited positive price, so the closed continuation has no positive terminal steady state.

**The immediate decision is narrow:** present the closed path as a finite-horizon benchmark with an explicit open sensitivity, or change the model and calibration enough to impose a closed-root requirement. Reprofiling the existing fertility slope is not sufficient.

Policy analysis comes only after this equilibrium choice is settled.